The Impact of BLM Protests on Policing and Crime

By Bobby Squires*

George Floyd's murder by police prompted a nationwide surge in protests during the summer of 2020. I leverage variation in initial protest timing and heterogeneity in officer injuries at protests to identify how shocks to the local police-public relationship affect proactive policing. Cities where officers are exposed to salient public backlash via peer injuries experience stronger depolicing, with a 25% long-run decline in officer-initiated calls. I also find evidence that civilians report low-level crime less frequently after protests.

I. Introduction

In the days following George Floyd's murder on May 25th, 2020, demonstrations against police brutality and racial discrimination broke out in virtually all major U.S. cities. Demands for police reform coalesced around Black Lives Matter (BLM),[1] which over the summer of 2020 grew into the single largest protest movement in U.S. history (Buchanan, Bui and Patel 2020). Media coverage of demonstrations during this period strained the police-public relationship by highlighting clashes between law enforcement and crowds (Mitchell et al. 2020). In a social climate defined by outrage over police violence, BLM protests brought civilians and officers into direct contact.

Floyd's death and its aftermath are not without precedent. Many of the largest protests in American history began in response to police violence (Cunningham and Gillezeau 2018, Ang and Tebes 2024). A central focus of the literature on police violence has been testing whether officers respond to backlash by depolicing —that is, reducing proactive crime prevention efforts —and if so whether this incentivizes crime (Owens 2019). An increase in crime following protests against police violence is referred to as a "Ferguson Effect," in reference to a supposed surge in violent crime following protests against the 2014 shooting of Michael Brown in Ferguson, Missouri (Mac Donald 2015).

Despite recent evidence for depolicing in the aftermath of police violence (Mikdash and Zaiour 2022, Rivera and Ba 2026, Premkumar 2025, Cheng and Long 2022), no existing work evaluates depolicing in the context of the nationwide BLM

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[1] Throughout this paper, BLM refers to the broad social movement against racial discrimination and police violence, rather than any particular organization or group.

protests following Floyd's death, the scope and scale of which were unique. This paper assesses the effect of public outrage at post-Floyd BLM protests on proactive policing, reporting, and crime.

I obtain reported crime data from Computer-Aided Dispatch (CAD) systems, drawing on a combination of city-specific open data portals, individual Freedom of Information Act requests, and the Open Policing Data project (Sowd and Otto 2025). I aggregate incident-level administrative datasets to obtain a daily panel of civilian 911 calls for service and officer-initiated calls from 40 U.S. cities[2] in 2020. For these cities, I construct a daily panel of protest activity during 2020 using the Crowd Counting Consortium's (CCC) dataset, which collects information on protests by web-scraping news articles, organizing websites, and social media (Chenoweth, Pressman and Ulfelder 2025).

Drawing on evidence that officers change their behavior at work when peers are injured on duty (Holz, Rivera and Ba 2023), I use variation in initial protest timing and officer injuries at protests to identify the dynamic effects of increased exposure to public outrage on proactive policing, civilian reporting, and reported crime. I control for weather effects with temperature and precipitation measures (Bodenhorn 2026) as well as pandemic confounders with weekly COVID-19 cases and daily mobility data (Ang et al. 2025).

Using a staggered difference-in-differences design, I find that the daily volume of officer-initiated calls declines by 25% in the months following a post-Floyd BLM protest with officer injuries, while civilian calls fall by roughly 8%. Low-priority disorder-related calls initially decline, but the effect does not persist. Violent crime-related calls remain stable through the sample period. Officers reduce proactive policing after exposure to civilian outrage, but I find no evidence of a subsequent increase in reported crime.

To test whether lower reporting rates may be masking underlying shifts in crime I conduct a similar analysis on a subsample of 9 cities with ShotSpotter systems.[3] ShotSpotter acoustically detects gunshots, and so serves as a benchmark for crime holding reporting constant. The ratio of civilian calls to ShotSpotter activations (the call-to-shot ratio), which roughly captures reports per gunshot, is naturally robust to changes in underlying crime, and has been shown to correlate strongly with local attitudes towards police (Ang et al. 2025). For cities where an officer sustains an injury at a protest, the civilian call-to-shot ratio drops by an average

[2]Data come from the following cities: Gilbert, AZ; Mesa, AZ; Phoenix, AZ; Fresno, CA; Los Angeles, CA; Menlo Park, CA; Richmond, CA; Sacramento, CA; San Diego, CA; San Francisco, CA; San Jose, CA; Stockton, CA; Fort Lauderdale, FL; Boise, ID; Bloomington, IN; Griffith, IN; St. John, IN; New Orleans, LA; Baltimore, MD; Cambridge, MA; Pittsfield, MA; Worcester, MA; Detroit, MI; Minneapolis, MN; Lincoln, NE; Las Vegas, NV; Manchester, NH; Nashua, NH; New York City, NY; Asheville, NC; Durham, NC; Cincinnati, OH; Portland, OR; Providence, RI; Charleston, SC; Nashville, TN; Austin, TX; Virginia Beach, VA; Seattle, WA; Tacoma, WA.

[3]ShotSpotter systems are designed to detect and pinpoint the location of gunshots automatically for rapid police response. Cities with ShotSpotter have specialized microphone systems installed at selected locations, and these locations do not vary over time. The subsample includes data from the following cities: Richmond, CA; Sacramento, CA; San Diego, CA; San Francisco, CA; Baltimore, MD; Pittsfield, MA; Minneapolis, MN; Cincinnati, OH; New York City, NY.

of 200 calls per gunshot over the rest of the year. This suggests marginal reporters avoided calling the police after protests in cities where unrest was most intense. This non-reporting effect may muffle the impact of a shift in underlying crime on reported crime.

This paper builds on existing work in two ways. First, I use a novel treatment framework—officer injuries proxy for police exposure to outrage—to exploit heterogeneity across cities where previous papers have struggled to find variation. This enables the identification of depolicing effects in the context of the largest episode of backlash against law enforcement in U.S. history (Buchanan, Bui and Patel 2020).

Additionally, by showing that reporting falls for cities where protests are most intense, I contribute to a growing body of work documenting the effects of a strained public-police relationship on crime reporting. This finding underscores the importance of distinguishing shifts in police oversight that result from public scandals, which may cause non-reporting, from non-publicized changes in supervision (for more see Rivera and Ba (2026)). The takeaway for policymakers is clear: a deteriorating police-public relationship can cause non-reporting. Therefore, building trust is critical to the proper function of law enforcement.

The rest of this paper is organized as follows. Section II surveys the relevant literature on police violence, depolicing, and reporting. Section III describes the data and gives background on CAD systems. Section IV gives the identification strategy and econometric model. Section V discusses results and concludes with policy implications.

II. Literature Review

A. Depolicing in Response to Backlash

Recent scholarship provides convincing evidence that proactive policing declines following public backlash against police violence (Mikdash and Zaiour 2022, Premkumar 2025, Rivera and Ba 2026). Arrests for low-level disorder offenses decline more than violent offenses, indicating that low-level offenses are more sensitive to proactive policing (Cheng and Long 2022, Premkumar 2025). Incidents of police violence with more coverage have a greater effect on police effort and therefore crime (Mikdash and Zaiour 2022, Premkumar 2025, Rivera and Ba 2026).

Premkumar (2025) develops a model of officer utility to explain depolicing, then tests predictions using monthly FBI Uniform Crime Reporting (UCR) data on arrests, Bureau of Justice Statistics National Crime Victimization Survey (NCVS) data on crime victimization, and a comprehensive dataset of officer-involved shootings and their coverage in online media from 2005 to 2016. Consistent with their model's predictions, they find that low-level arrests decline following officer-involved shootings while arrests for violent offenses remain unchanged. In contrast, the volume of low-level crimes rises modestly, while the volume of

violent offenses increases sharply. The insights from the model in Premkumar (2025), that depolicing after public backlash manifests in reduced effort for low-level offenses, match intuition and are corroborated in a qualitative study that interviewed police (Gau, Paoline and Paul 2022).

Cheng and Long (2022) use a CAD dataset from St. Louis to directly study officer-initiated calls and arrests in majority Black neighborhoods following police shootings in other cities from 2014 to 2016. They also find a substantial drop in officer-initiated calls for Black neighborhoods, as well as a decline in arrests concentrated in low-level offenses. Similarly Mikdash and Zaiour (2022) documentes declines in low-level arrests following non-publicized police shootings in Minneapolis driven by a decline in officer-initiated calls (Mikdash and Zaiour 2022). Though not its main contribution, Rivera and Ba (2026) shows that police reduce proactive efforts and crime rises following a major police scandal in Chicago.

Existing work clearly shows depolicing is a persistent phenomenon following public outrage over police violence.[4] The intuitive conceptual framework, that officers facing public backlash are less willing to initiate a marginal public engagement, fits theoretical and empirical models of depolicing well. Differences in the magnitude of depolicing across crime types make sense within this framework, and are supported by the data. Facing public outrage, officers reduce marginal policing efforts. This drives down low-level arrests temporarily, while violent arrests remain stable since police are attentive to high-priority crime.

B. Crime and Reporting Following Police Violence

The literature discussed above has focused on arrests and officer-initiated calls. These measures provide information about the scale of depolicing, and for what types of offenses it is occurring, but speak little to how crime itself may have shifted as a result of depolicing. Even if depolicing is shown to occur, the question remains: is there a "Ferguson Effect?"

A natural approach to measuring crime following depolicing would be to look at longer run changes in the volume of civilian calls for service. But calls for service only capture *reported* crime which is necessarily a function of civilian trust in law enforcement. This means that in cities where depolicing occurs as a result of public outrage, shifts in reported crime after a depolicing episode may be contaminated by changes in civilian reporting, and not align with the predictions of depolicing theory: that crime rises after a reduction in proactive policing. Following an officer-involved shooting, civilians reduce calls due to lost trust, while officers reduce proactive policing. The latter may drive up underlying crime, canceling out reporting effects.

Earlier work showed that 911 calls in Milwaukee decreased following the beating

[4]It should be noted that empirical work on the historical anti-policing protests of the 1960s and 70s shows the opposite, officers respond by cracking down and using greater force (Cunningham and Gillezeau 2021, Cunningham and Gillezeau 2018). Depolicing appears to be a 21st century phenomenon.

of Frank Jude by police officers (Desmond, Papachristos and Kirk 2020). This directly conflicts with the effects one would expect if depolicing took place, which would incentivize marginal offenders, leading civilians to demand police service more than before the incident. The idea that civilians who are exposed to news about a police killing will be less likely to call 911 is intuitive. Studies examining changes in the volume of 911 calls following similar incidents of police misconduct have had mixed results, with some finding no effect (Cohen et al. 2019) and others finding differing results within and between cities (Brantingham, Mohler and MacDonald 2022). It seems likely this is due to a failure to account for depolicing effects on underlying crime in reporting studies.

Two recent studies address the measurement of civilian reporting tendencies, leveraging ShotSpotter data to identify changes in reporting independently of changes in crime. Using the call-to-shot ratio, Ang et al. (2025) find that George Floyd's murder caused a significant drop in reporting across a sample of several U.S. cities. They additionally show that the call-to-shot ratio is highly correlated with survey-based measures of community-level trust (Ang et al. 2025). Mikdash and Zaiour (2024) leverage ShotSpotter activations and granular call location measurements to test changes in reporting after use-of-force cases. They find a decrease in call-to-shot ratios following use-of-force incidents, with the greatest effects in minority communities (Mikdash and Zaiour 2024). Importantly, this change is driven by both increases in gun crime and decreases in reports. These findings highlight the importance of accounting for how changes in community trust impact reported crime when trying to estimate the effects of depolicing on reported crime.

Rivera and Ba (2026) provides a compelling framework for understanding how depolicing and reported crime interact following an incident of police violence. They separate effects of police shootings on reported crime into those attributable to officers responding to increased oversight and those that result from public outrage. Then using events in which police oversight increased with no corresponding public outrage, and comparing them with those where police oversight increased following a prominent officer-involved shooting, they identify the effects of a pure increase in oversight on crime, and isolate them from the effects of public backlash against police. Their results are consistent with their conceptual framework: greater scrutiny resulting from public scandal causes an increase in crime due to depolicing and non-reporting incentivizing crime simultaneously (Rivera and Ba 2026).

Even when depolicing can conclusively be shown to occur, such as through a decline in officer-initiated calls, attributing any changes in crime to depolicing is difficult. As the literature has shown, observed crime is a function of both underlying crime and civilian reporting, both of which respond to the same public outrage that induces depolicing. This paper builds a clearer picture of how depolicing impacts crime by separating changes in violent crime (which is less sensitive to depolicing), changes in disorder (which is more sensitive), and changes

in reporting behavior using the call-to-shot ratio.

III. Data

A. Computer-Aided Dispatch Background and Data

Police departments are called to address a broad range of incidents on any given day, most of which never result in any arrest. Regardless of the reason for the call, civilian attempts to contact police are referred to as calls for service (CFS) (Bureau of Justice Statistics 2022). Dispatchers serve as intermediaries between officers and civilians requesting service. In order to efficiently respond to calls for service, dispatchers use Computer-Aided Dispatch (CAD) systems to communicate with officers who are on patrol or responding to a call. Field officers let dispatch know through CAD systems when they initiate an encounter so that dispatchers know not to assign them to incoming calls nearby. Each observation in a CAD database includes the time, date, location, and reason for a call. These observations encompass civilian calls for service and officer-initiated encounters (Bureau of Justice Statistics 2022). Both are recorded as *calls* in CAD parlance: a call record is an entry in the dispatch log, not necessarily a telephone call. An officer-initiated record is created when an officer notifies dispatch that they have begun an encounter, rather than being dispatched to one. I follow the convention in the CAD literature (Cheng and Long 2022, Mikdash and Zaiour 2022) and refer to these records throughout as officer-initiated calls. Given the essential nature of CAD systems for the function of law enforcement agencies, as well as legal requirements pertaining to administrative record keeping, CAD datasets can be reliably taken to represent the universe of civilian calls and officer-initiated calls within their respective cities.

To measure proactive police engagements and reported crime, I obtain incident-level CAD datasets from 40 U.S. cities,[5] drawing on a combination of city-specific open data portals,[6] individual Freedom of Information Act requests, and the Open Policing Data project (Sowd and Otto 2025). Within each city's CAD dataset, observations contain the incident date and a department-specific call type variable assigned either automatically (for some alarms and some officer-initiated calls) or by the dispatcher. Call descriptions vary considerably across departments and states, so standardizing call types is necessary to make comparisons across cities.

I compile a list of all raw call type labels for every department in the dataset, then give an AI language model (Anthropic 2026) written guidelines to determine whether each call type falls into four non-exclusive categories,[7] as well as a human-

[5]Daily coverage is complete for 35 of the 40 cities. Four are missing a single day. Manchester, New Hampshire is missing 43 days, from November 19 through December 31, 2020. Manchester never experiences an officer-injury protest, so this gap affects only the never-treated comparison group, and only on calendar dates far outside the 14- and 30-day event windows.

[6](City of Detroit Open Data Portal 2026, City of Sacramento Public Saftey Data 2026, Open Data Cincinnati 2026, NYC Open Data 2026)

[7]The text instructions given to Claude Code for call classification are included with replication package on my website. Spot checks were performed on the categorizations to ensure labeling accuracy.

labelled sample of call types to train with. These categories are: violent-crime-related incidents, low-level public-order-related incidents (or "disorder" calls), ShotSpotter activations, and officer-initiated calls. The results of the models classification are recorded in a pooled registry of all call types. I categorize observations as violent if the description variable references any FBI UCR violent offenses: murder, nonnegligent manslaughter, rape, robbery, or aggravated assault (Federal Bureau of Investigation 2018). I categorize observations as disorder calls if they reference *visible* quality-of-life concerns. For example, observations that reference homelessness or public drug use are categorized as public order calls, but observations that reference drug use with no mention of locale are not categorized as public order calls, since the call may not have been made for a public incident. Similarly, officer-initiated call types reference common police descriptions of self-initiated activity such as "directed patrol" or "Mobile Data Computer (MDC)". Civilian calls are defined as any non-officer-initiated call.

After categorizing, I collapse each dataset to get daily counts of calls overall and by category, then construct a city-day level panel of call data. I combine this with Census population data for each city. Table 1 reports summary statistics for the CAD panel. Figure 1 shows trends in average call rates by type throughout 2020.

Of the 40 cities in the sample, 11 cities include ShotSpotter activations automatically in their CAD dataset. The lack of ShotSpotter data elsewhere is not wholly surprising since ShotSpotter is controversial and expensive. Cities with no ShotSpotter observations throughout the sample period have all observations marked missing and are excluded from subsample analysis using ShotSpotter data. One of the 11 (Fort Lauderdale, Florida) records its first activation only in October, months after the murder of George Floyd, so it has no usable pre-treatment period and is dropped. Menlo Park, California records activations on only six days in all of 2020, too little signal in the denominator for its call-to-shot ratios to be meaningful, so it is dropped as well. The ShotSpotter subsample is therefore the remaining 9 cities, covering 3,293 city-days. Table 2 reports summary statistics and Figure 2 shows trends in average call rates by type for the ShotSpotter subsample. Figure 3 shows trends in ShotSpotter-specific outcomes.

B. Weather and COVID-19 Controls

Because hotter, drier weather increases crime (Bodenhorn 2026), I include daily temperature averages and precipitation totals as weather controls, obtained using Open-Meteo's API (Zippenfenig 2023). Given the unique context of the post-Floyd BLM protests in 2020, it is necessary to control for the pandemic's effects on civilian mobility and the police force. Following Ang et al. (2025), I use mobility data from Google's COVID-19 Community Mobility reports to control for variation in public foot traffic and time spent at home. The mobility variables are percentage changes in foot traffic (measured with location tracking information from smartphones) at retailers and residential areas relative to their pre-pandemic

Table 1—Summary Statistics: Daily Calls for Service and Controls — Full Sample

	Obs.	Mean	Std. Dev.	p25	Median	p75	Min	Max
Calls for Service								
Officer-initiated calls (per 100,000)	14,593	43.6	65.2	4.4	18.7	57.5	0.0	521.4
Civilian-initiated calls (per 100,000)	14,593	174.0	99.9	107.0	145.4	215.8	0.4	776.3
Violent calls (per 100,000)	14,593	4.3	5.6	0.8	2.4	5.1	0.0	57.5
Disorder calls (per 100,000)	14,593	36.6	22.1	21.9	32.6	47.2	0.0	215.8
Controls								
Population (thousands)	14,593	720.3	1,451.6	150.6	320.8	652.5	16.5	8,805.6
New COVID-19 cases	13,753	1,837.6	5,164.4	78.0	436.0	1,415.0	0.0	104,016.0
Temperature (°C)	14,593	15.3	9.4	8.7	15.4	22.4	-22.2	39.8
Precipitation (mm)	14,593	2.5	6.8	0.0	0.0	1.3	0.0	155.9
Retail mobility	12,776	-24.3	18.0	-35.0	-24.0	-13.0	-92.0	87.0
Residential mobility	12,792	11.6	7.4	7.0	11.0	16.0	-5.0	37.0

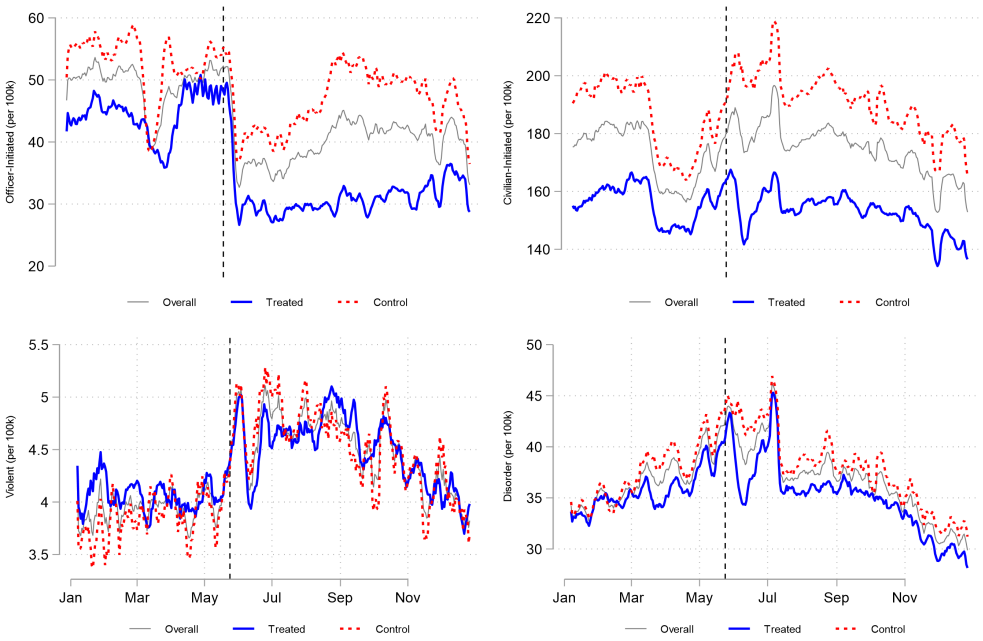
Note: Observations are at the city-day level (N $\approx$ 14,640 city-days = 366 days $\times$ 40 cities). Due to the leap year, some CAD datasets contain 366 observations while others contain 365. There are approximately 20 missing days due to known CAD outages in 2 cities. Google Mobility indices are percentage changes in foot traffic relative to the pre-pandemic period. Google Mobility and COVID-19 case data are missing prior to the pandemic.

Table 2—Summary Statistics: Daily Calls for Service and Controls — ShotSpotter Subsample

	Obs.	Mean	Std. Dev.	p25	Median	p75	Min	Max
Calls for Service								
Officer-initiated calls (per 100,000)	3,293	65.6	98.6	14.8	28.8	62.0	0.0	521.4
Civilian-initiated calls (per 100,000)	3,293	228.7	104.7	144.5	186.5	325.6	0.4	776.3
Violent calls (per 100,000)	3,293	6.3	6.1	2.6	4.0	9.5	0.0	57.5
Disorder calls (per 100,000)	3,293	48.7	22.0	33.3	47.8	60.8	0.0	215.8
ShotSpotter activations (per 100,000)	3,293	2.0	3.2	0.1	0.7	2.3	0.0	30.5
Officer calls per shot	3,286	101.2	158.8	9.0	35.6	135.0	0.1	2,370.0
Civilian calls per shot	3,286	347.9	403.3	82.0	192.5	430.0	6.8	2,321.0
Controls								
Population (thousands)	3,293	1,453.5	2,628.3	309.6	524.9	878.4	44.0	8,805.6
New COVID-19 cases	3,104	1,887.8	4,548.8	48.0	542.0	1,655.0	0.0	37,059.0
Temperature (°C)	3,293	13.4	8.5	8.7	14.0	19.7	-22.2	35.2
Precipitation (mm)	3,293	2.2	6.3	0.0	0.0	0.9	0.0	133.5
Retail mobility	2,871	-30.2	19.2	-42.0	-29.0	-20.0	-92.0	20.0
Residential mobility	2,888	12.9	8.0	7.0	13.0	17.0	-3.0	37.0

Note: Restricted to the 9-city ShotSpotter subsample. Ratios are defined on days with no ShotSpotter activations by adding 1 to both the numerator and denominator. Observations are at the city-day level (N $\approx$ 3,293 city-days $\approx$ 366 days $\times$ 9 cities). Due to the leap year, some CAD datasets contain 366 observations while others contain 365. There are approximately 20 missing days due to known CAD outages in 2 cities. Google Mobility indices are percentage changes in foot traffic relative to the pre-pandemic period. Google Mobility and COVID-19 case data are missing prior to the pandemic.

FIGURE 1. 7-DAY MOVING AVERAGE OF CALL RATES — FULL SAMPLE



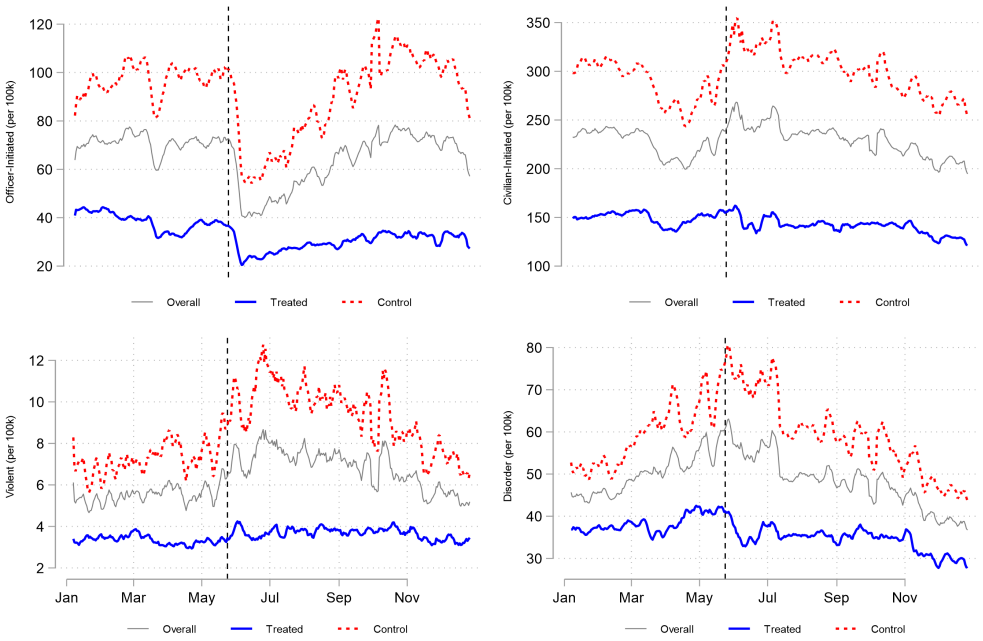
Note: Seven-day trailing moving average of daily call rates per 100,000 residents. Treated cities are those with at least one BLM protest resulting in officer injuries after May 25, 2020. Control cities had BLM protest activity but no officer injuries. Dashed vertical line marks George Floyd's death (May 25, 2020).

average. Additionally, I include weekly Centers for Disease Control data on new COVID-19 cases to measure the spread of the virus. Sick officers are likely to reduce effort at work or call out, leading to strain on the police force and lower officer-initiated calls. Civilians who are incapacitated are likely to stay home and therefore report less crime. Marginal offenders may not commit crimes when sick. Tables 1 and 2 report summary statistics on controls for the main sample and the ShotSpotter subsample.

C. Protests

To measure protest activity, I use the Crowd Counting Consortium's (CCC) database of all protests in the U.S. during 2020. Data were collected by web-scraping news sites and articles, protest organizing websites, and social media (Chenoweth, Pressman and Ulfelder 2025). There was also an option for members of the public to submit information about a protest, which was vetted manually. While it is possible that the CCC protest dataset does not contain the complete universe of protests during 2020, the method of collection ensures that major protests, that is, those that received broad media coverage, are included. As

FIGURE 2. 7-DAY MOVING AVERAGE OF CALL RATES — SHOTSPOTTER SUBSAMPLE



Note: Same as Figure 1, restricted to the 9-city ShotSpotter subsample.

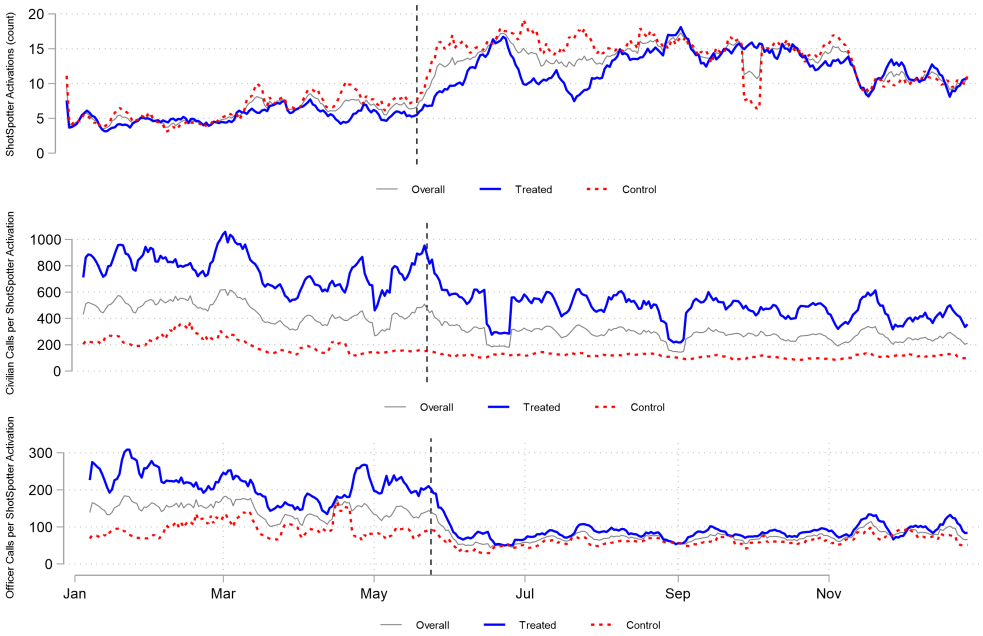
mentioned previously, post-Floyd BLM protests were the largest and most widely covered in U.S. history (Buchanan, Bui and Patel 2020, Heaney 2020). CCC data are therefore a reliable measure of all the protests of interest for this analysis, which focuses on major instances of public backlash against police during the BLM movement.

Each entry in the raw CCC dataset represents a single protest. Observations include protest date, time, city, state, the demands of the protesters, an indicator for officer injury at the protest, and other variables not included here. First, I identify BLM protests using a text search on protest demands for topics related to policing or race.[8] I restrict to only protests in the 40 cities of interest, then collapse to a daily city-level panel of protest activity, with variables for the number of protests, the number of BLM protests, and the number of BLM protests where an officer was injured. Summary statistics on protests for the main sample and the ShotSpotter subsample are reported in Tables 3 and 4 respectively.

One concern with the dataset is that it may not accurately record all officer injuries at protests. Again, the methods of data collection and the scope of media coverage during BLM protests are reassuring. Officer and crowd injuries

[8]Specifically, BLM protests are defined as those whose claims contain the following keywords: "police brutality", "Black Lives Matter","racism","white supremacy","George Floyd", or "policing".

Figure 3. 7-Day Moving Average of ShotSpotter Activity — ShotSpotter Subsample



Note: Seven-day trailing moving average of ShotSpotter-related outcomes for the 9-city ShotSpotter subsample. Call-to-shot ratios add one to both the numerator and denominator. Dashed vertical line marks George Floyd's death (May 25, 2020).

Table 3—Summary Statistics: Protest Activity — Full Sample

	Nonzero Days	Total Protests	Mean	Std. Dev.	Median	p95	Max
Total protests	2,764	5,393	1.95	2.52	1	5	41
BLM protests	1,796	3,396	1.89	2.68	1	4	41
Protests with officer injuries	71	92	1.30	1.10	1	2	9
BLM protests with officer injuries	66	85	1.29	1.13	1	2	9

Note: Observations are at the city-day level (N $\approx$ 14,640 city-days = 366 days $\times$ 40 cities). Due to the leap year, some CAD datasets contain 366 observations while others contain 365. Zero-protest days are excluded here for summary purposes, but included in the analysis. All statistics in the above table are computed conditional on at least one protest of that type occurring.

Table 4—Summary Statistics: Protest Activity — ShotSpotter Subsample

	Nonzero Days	Total Protests	Mean	Std. Dev.	Median	p95	Max
Total protests	848	2,471	2.91	4.06	1	9	41
BLM protests	556	1,645	2.96	4.41	2	11	41
Protests with officer injuries	15	29	1.93	2.22	1	9	9
BLM protests with officer injuries	14	26	1.86	2.32	1	9	9

Note: Restricted to the 9-city ShotSpotter subsample. Observations are at the city-day level (N $\approx$ 3,293 city-days $\approx$ 366 days $\times$ 9 cities). Due to the leap year, some CAD datasets contain 366 observations while others contain 365. Zero-protest days are excluded here for summary purposes, but included in the analysis. All statistics in the above table are computed conditional on at least one protest of that type occurring.

were widely reported during 2020 as liberal and conservative media advanced narratives of protester or officer violence. It seems highly likely that officer injuries which elicit peer responses are also those reported by the news, and that cities where officers experience the most intense backlash are those where such injuries occurred at protests.

IV. Empirical Strategy

A. *Predictions Based on Depolicing Theory*

Depolicing theory (Owens 2019, Premkumar 2025) suggests that officers will reduce crime-fighting efforts as their relationship with the communities they serve deteriorates. Empirical work shows that both arrests for low-level offenses and officer-initiated calls fall after police violence (Mikdash and Zaiour 2022, Cheng and Long 2022, Premkumar 2025, Rivera and Ba 2026). The magnitude of the decline is greater for highly publicized police misconduct (Mikdash and Zaiour 2022, Premkumar 2025, Rivera and Ba 2026). A separate but related literature shows that civilians respond to police violence by reducing discretionary reporting (Ang et al. 2025, Mikdash and Zaiour 2024, Desmond, Papachristos and Kirk 2020). When these two changes interact in a highly publicized instance of police violence, theoretical and conceptual frameworks suggest that the combination of lower reporting by civilians and less proactive crime prevention by officers will produce a decline in reported crime in the short-term, with more underlying crime (particularly for low-level offenses) in the long-term.

Whether this proposal holds true is difficult to test for a number of reasons. First, reported crime necessarily depends on civilian attitudes towards the police, and measuring underlying crime at high frequency is difficult. If reported crime shifts following backlash, it is not clear through which channel the shift occurred: reporting or crime. Unreported crime estimates, when they are available, are typically low-frequncy (as in NCVS surveys), making it impossible to confirm whether depolicing has occurred.

This paper aims to overcome these challenges, by first testing whether exposure

to outrage causes depolicing at the city level, then examining whether reporting also declines following outrage. Changes in these measures combined with short- and long-term changes in various types of reported crime are suggestive of underlying crime dynamics.

B. Identification Strategy and Econometric Model

Using a staggered difference-in-differences design (Miller 2023), I estimate the effects of police exposure to outrage on proactive policing, civilian reporting, and crime in the days following a post-Floyd BLM protest. All 40 cities in the sample experience a BLM protest after the death of George Floyd, but only 17 cities have protests with officer injuries. Thus, I consider a city treated following the first post-Floyd BLM protest[9] with an officer injury, arguing that an officer injury is as good as randomly assigned to cities conditional on protest, and is a good proxy for greater backlash against police. I leverage variation in protest timing in order to estimate the dynamic effects of outrage, accounting for heterogeneity across treatment cohorts using the interaction-weighted estimator proposed by Sun and Abraham (2021).

My interpretation of the treatment draws on Holz, Rivera and Ba (2023), which finds that officers change use-of-force behavior when their peers in different departments sustain an on-duty injury. I study a more direct social link, focusing on the response of officers in the department where the injury occurred. I assume that peer injuries are understood by other officers as a symptom of public backlash. Figure 4 shows the timing of protests following Floyd's death and Figure 5 shows treatment uptake by cohort. Figure 6 shows the same for the ShotSpotter subsample.

It is well known that the standard two-way fixed effects estimator is biased in settings with staggered adoption unless treatment effects are assumed to be homogeneous across cohorts. Heterogeneity across cohorts can lead to invalid comparisons of earlier-treated cities with later-treated cities (Goodman-Bacon 2021). To account for this, I use the interaction-weighted estimator, which recovers a weighted average of the cohort treatment effects (Sun and Abraham 2021). I use the interaction-weighted estimator in both a dynamic and collapsed design (where the post-treatment period is a single interaction-weighted coefficient). Formally, I estimate the following model is the following:

$$Y_{i,t} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - g_i = k] + \theta_i + \gamma_t + \mathbf{X}'_{it}\alpha + \epsilon_{i,t}$$

where $Y_{i,t}$ is the logged count of calls *or* call-to-ShotSpotter ratios for city i on day t, and g_i is the treatment date for city i. The indicator $\mathbf{1}[t - g_i = k]$ is 1 when an observation for city i is k days relative to its treatment day g_i. The

[9]Hereafter, since all protests of interest occur after the death of George Floyd and are related to BLM, I simply refer to them as protests.

β_k represent the treatment effects: percentage change in daily call volume or changes in the call-to-ShotSpotter ratio, at event time k with $k = -1$ used as the reference period.[10] The θ_i and γ_t are city and time fixed effects respectively and $\mathbf{X}_{i,t}$ is a vector of time-varying city-specific controls including new COVID-19 cases, temperature, precipitation, and mobility indices (Bodenhorn 2026, Ang et al. 2025). Standard errors are clustered at the city level.

Tests for differences in pre-treatment means (shown in Tables 5 and 6) show that treatment and control groups differ in daily call counts and population. These level differences are absorbed by city fixed effects in my difference-in-differences design. Adjusting for population shows that raw counts of calls overstate diff differences between treatment and control groups, whose call rates per 100,000 people are indistinguishable. Pre-treatment retail mobility differs across treatment and control groups, but its inclusion as a control assuages concerns. Average protests differ between treatment and control groups as well, but importantly BLM protests do not. Treatment and control cities do not differ in most other observable characteristics.

The central assumptions of my staggered difference-in-differences model are threefold. The first assumption is that treatment and control cities exhibit parallel trends in the absence of intervention. Figures 1, 2, and 3 suggest this holds true. Call-to-shot ratios and call rates for treatment and control groups appear to track each other strongly in the pre-Floyd period. My controls account for the effects of the pandemic and weather on cities, strengthening this assumption.

Secondly, the model assumes that there is no anticipatory effect. A general lack of pre-treatment effects in my event study plots and tables also supports this. I perform F-tests for differences in variance on my pre-treatment event study coefficients and find them largely insignificant. Additionally, time fixed effects capture post-Floyd pre-event depolicing as a common shock across cities.

Lastly, the Stable Unit Treatment Value Assumption (SUTVA) mandates that treatment of one unit have no effect on other units, and that treatment be uniform across cities. Restricting the sample to cities that experience protests reduces concerns that reactions to peer injuries in other departments may have contaminated control cities, since all departments had perceptions shaped primarily by their local protests. While protest intensity may vary within the treatment group, officer injuries were generally rare at protests, so the impact of an additional injury on officers is likely diminishing. In other words, the officer-perceived backlash resulting from two peer injuries as opposed to one is less tangible than the difference between one injury and none.

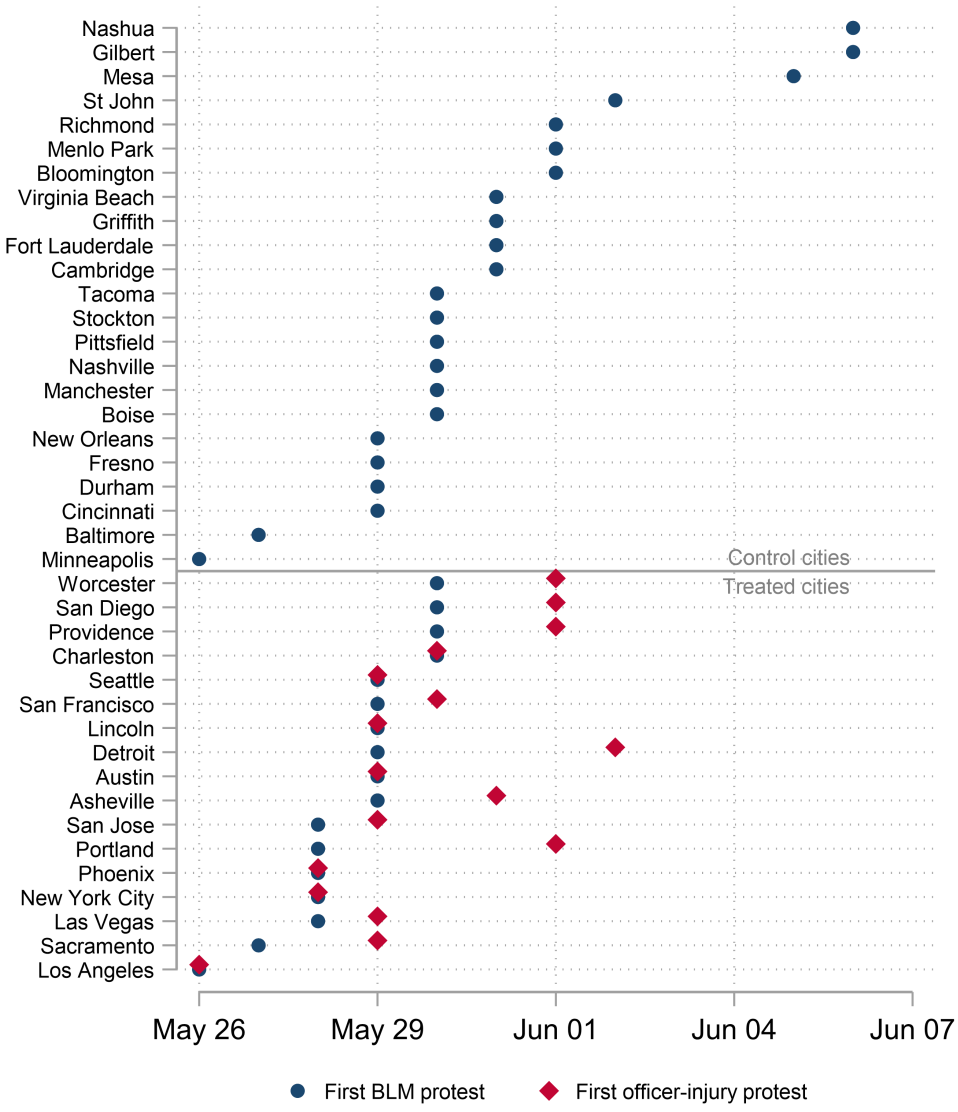
[10] Since the main specification is a 2-day pooled regression the reference period becomes $k \in \{-1, -2\}$. In the collapssed design time dummies collapse to pre- and post-treatment. The standard regression equation above is given for clarity.

Table 5—Pre-Treatment Balance: Treated vs. Control Cities

	Control	Treated	Difference	p-value
Daily Call Counts				
Officer-initiated calls (count)	183.91	765.38	-581.47	0.131
Civilian-initiated calls (count)	559.01	1,750.41	-1,191.40	0.047
Violent calls (count)	14.51	50.24	-35.73	0.026
Disorder calls (count)	111.32	361.81	-250.49	0.008
Daily Call Rates (per 100,000)				
Officer-initiated calls (per 100,000)	52.88	44.56	8.31	0.687
Civilian-initiated calls (per 100,000)	185.96	156.42	29.55	0.353
Violent calls (per 100,000)	3.89	4.09	-0.20	0.898
Disorder calls (per 100,000)	37.69	35.82	1.87	0.788
Controls				
Population (thousands)	263.11	1,334.44	-1,071.33	0.020
New COVID-19 cases	199.91	1,021.92	-822.00	0.143
Temperature (°C)	9.83	10.95	-1.12	0.536
Precipitation (mm)	2.72	2.49	0.22	0.604
Retail mobility	-25.79	-30.78	4.99	0.034
Residential mobility	12.24	13.79	-1.55	0.112
Protest Activity (daily avg.)				
Total protests	0.04	0.19	-0.15	0.000
BLM protests	0.01	0.01	-0.01	0.539

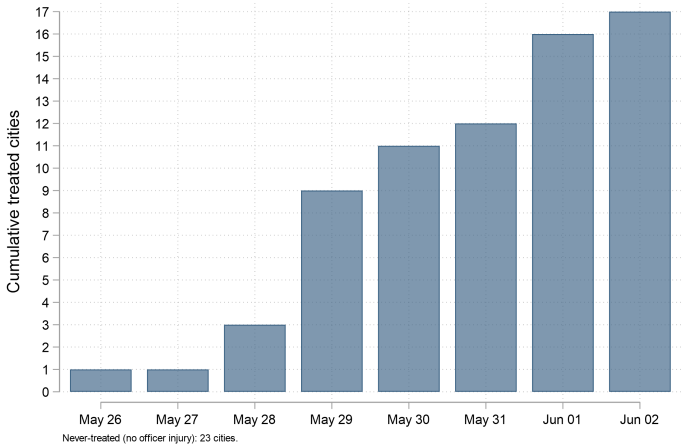
Note: Pre-treatment means from January 1–May 24, 2020, collapsed to the city level. Treated cities are those that experience a BLM protest with officer injury (N = 17). Control cities are never-treated (N = 23). Difference is control minus treated. p-value from a two-sample t-test. Mobility indices are percentage changes in foot traffic relative to the pre-pandemic baseline.

FIGURE 4. TREATMENT TIMING — FULL SAMPLE



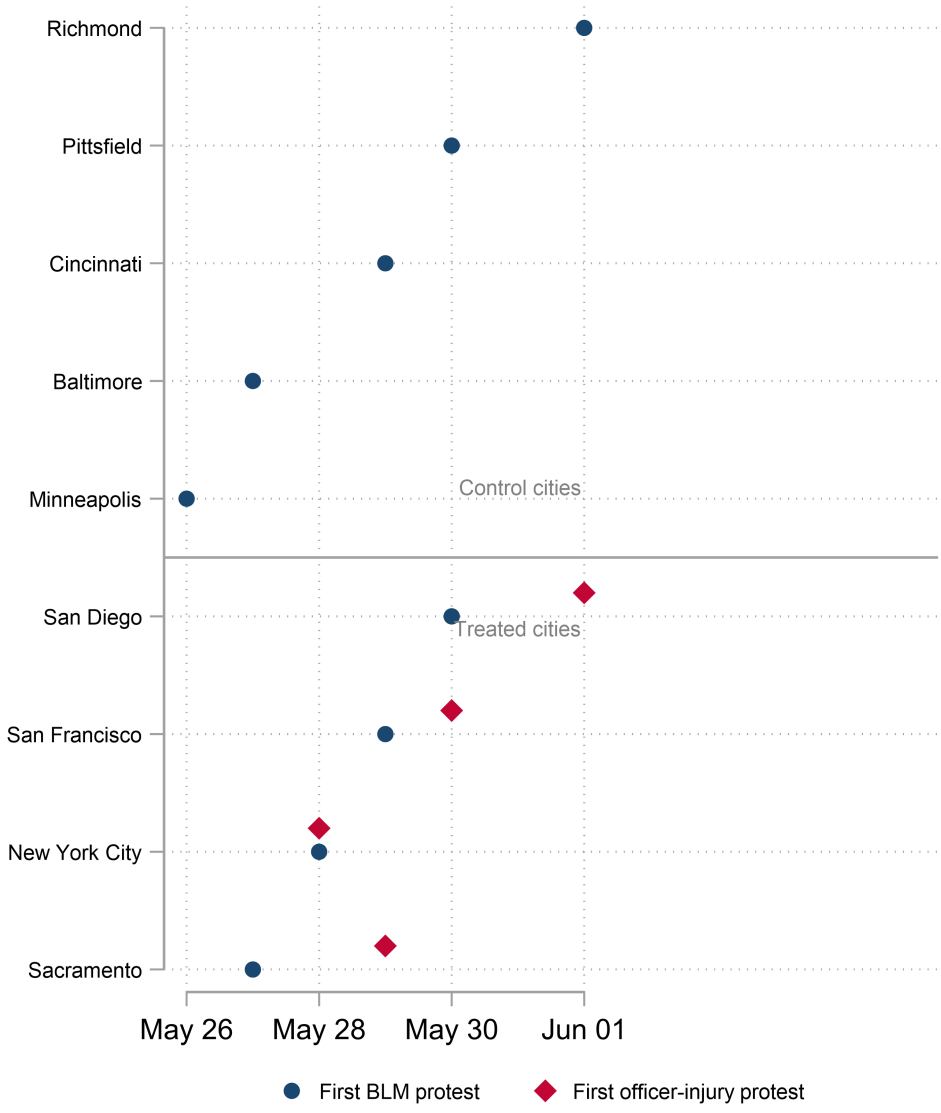
Note: Navy circles indicate the date of each city's first BLM protest after May 25, 2020. Cranberry diamonds indicate the date of the first BLM protest with officer injuries (the treatment date). Cities are sorted with treated cities below the dividing line and control cities (BLM protest activity but no officer injuries) above. Two cities with no BLM protest activity are excluded from the sample.

FIGURE 5. TREATMENT UPTAKE — FULL SAMPLE



Note: Bars show the cumulative number of cities that had received treatment (first BLM protest with officer injuries) by each date in the week following the George Floyd killing.

FIGURE 6. TREATMENT TIMING — SHOTSPOTTER SUBSAMPLE



Note: Same as Figure 4, restricted to the 9-city ShotSpotter subsample; Fort Lauderdale and Menlo Park are excluded (see Section III).

Table 6—Pre-Treatment Balance: Treated vs. Control Cities — ShotSpotter Subsample

	Control	Treated	Difference	p-value
Daily Call Counts				
Officer-initiated calls (count)	451.93	1,887.91	-1,435.98	0.379
Civilian-initiated calls (count)	1,001.41	3,853.37	-2,851.96	0.245
Violent calls (count)	29.50	79.37	-49.87	0.298
Disorder calls (count)	197.64	684.95	-487.31	0.136
Daily Call Rates (per 100,000)				
Officer-initiated calls (per 100,000)	96.85	38.48	58.37	0.404
Civilian-initiated calls (per 100,000)	289.84	150.11	139.72	0.025
Violent calls (per 100,000)	7.37	3.34	4.03	0.202
Disorder calls (per 100,000)	59.90	38.38	21.52	0.128
ShotSpotter				
ShotSpotter activations (per 100,000)	1.99	0.24	1.75	0.042
Officer calls per shot	93.92	210.53	-116.61	0.346
Civilian calls per shot	201.64	776.25	-574.61	0.006
Controls				
Population (thousands)	297.11	2,898.33	-2,601.23	0.179
New COVID-19 cases	147.49	2,884.65	-2,737.15	0.289
Temperature (°C)	6.28	11.59	-5.30	0.100
Precipitation (mm)	2.63	1.75	0.87	0.305
Retail mobility	-27.04	-38.82	11.78	0.054
Residential mobility	12.48	16.13	-3.66	0.136
Protest Activity (daily avg.)				
Total protests	0.07	0.28	-0.21	0.025
BLM protests	0.03	0.01	0.03	0.461

Note: Restricted to the 9-city ShotSpotter subsample. Pre-treatment means from January 1–May 24, 2020, collapsed to the city level. Treated cities are those that experience a BLM protest with officer injury (N = 4). Control cities are never-treated (N = 5). Difference is control minus treated. p-value from a two-sample t-test. Mobility indices are percentage changes in foot traffic relative to the pre-pandemic baseline.

V. Results and Conclusions

A. Collapsed Difference-in-Differences Results

Begining with a collapsed version of my model, where the post-treatment period has a single interaction-weighted coefficient, find that officer-initiated calls decline by 25% throughout the reaminder of the year following atreatment protest. Civilian-initiated calls also fall by about 8% (see Table 7). These results are consistent with a withdrawal of both proactive policing and discretionary civilian reporting in the immediate aftermath of outrage exposure. This is suggestive of a sustained pullback rather than a temporary disruption to individual call types, and matches the existing literature (Mikdash and Zaiour 2022, Premkumar 2025, Cheng and Long 2022) and my own predictions.

I find no significant changes in violent crime that persist through the post-treatment period consistent, with models of depolicing (Premkumar 2025) which suggests such a rise is temporary. I also find no changes in disorder crime that persist through the post period in the collapsed design. This broadly aligns with the model in Premkumar (2025), which anticipates that changes in reported low-level offenses may adjust dynamically to an increase in crime. An initial drop in low-level disorder offenses may be driven by a true drop in reporting and depolicing, but a subsequent rise in underlying crime, as marginal offenders adjust to a reduced probability of being caught, may make collapsed estimates unreliable. To test for changes in reporting that may be masking the single period post-treatment effect, I turn to the ShotSpotter subsample.

Table 7—Collapsed Difference-in-Differences — Full Sample

	Log Officer	Log Civilian	Log Violent	Log Disorder
Post × Treated	-0.252**	-0.0770***	0.00472	-0.0296
	(0.124)	(0.0280)	(0.0397)	(0.0385)

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Each column reports the coefficient on a single pooled post-treatment indicator ($t \geq 0$) from interaction-weighted estimates Sun and Abraham (2021), using never-treated cities as the comparison group; the reference period is days -2 to -1, as in Table 9. City and date fixed effects absorbed. The outcome is the log of the respective daily call count. Controls include daily mean temperature, precipitation, weekly new COVID-19 cases, and retail and residential mobility indices. Standard errors clustered at the city level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In the ShotSpotter subsample, the civilian call-to-shot ratio declines substantially, by about 200 calls per activation, following treatment (see Table 8). I include the officer-initiated calls per ShotSpotter activation as a secondary measure of depolicing. This declines by about 67 calls per activation, strengthening the depolicing argument in the subsample. As mentioned above, because ShotSpotter activations are independent of reporting behavior, this ratio captures a change

in the probability of reporting. The decline in this ratio therefore suggests that marginal callers in treated cities avoid contacting police following outrage, above and beyond any change in underlying crime. This is consistent with Ang et al. (2025) and Mikdash and Zaiour (2024), who document similar reporting declines following police violence. This suggests such reporting declines mask treatment effects in later periods.

Table 8—Collapsed Difference-in-Differences — ShotSpotter Subsample

	Log Officer	Log Civilian	Log Violent	Log Disorder	Officer/Shot	Civilian/Shot
Post × Treated	0.227	-0.0957***	-0.0415	0.0137	-66.78***	-199.9***
	(0.269)	(0.0332)	(0.0831)	(0.0601)	(23.93)	(23.02)

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Same as Table 7, restricted to the 9-city ShotSpotter subsample. Officer/Shot and Civilian/Shot are the officer and civilian call-to-shot ratios, dividing officer- or civilian-initiated calls by ShotSpotter activations, with +1 added to numerator and denominator.

B. Dynamic Effects

To examine dynamic treatment responses, I estimate event study coefficients using 14-day pre- and post-treatment windows (see Tables 9 and 10, Figures 7, 8, and 11). To reduce noise while preserving the shape of the treatment response, adjacent pairs of event-time days are averaged into 2-day pooled bins, so that each plotted coefficient represents the average effect over a two-day window (Miller 2023). I include an end-cap in all specifications, so the final coefficient represents the entire post-window period. Pre-period pooled coefficients are generally insignificant (F-test p-values are reported in figure notes), supporting the parallel trends assumption. Additionally, I detect no significant pre-treatment effects (at the 5% level) in any of the six main variables of interest (logged calls in the main sample, ShotSpotter ratios in the subsample) with this specification. I pair these with an alternative 30-day pre- and post-treatment window using 5-day pooled bins to test robustness to longer windows (see Figures 9, 10, and 12) and find broadly similar effects, though it should be noted this longer specification shows some pre-trends.

In the main specification for the full sample, disorder calls decline by 15% in the two days immediately following treatment but subsequently recover to pre-treatment levels within the estimation window. Any initial withdrawal of reported disorder is short-lived. Civilian calls fall significantly, by 7 to 19% throughout the entire window and into the post-window period. Combining this with the results from the ShotSpotter subsample, which shows a decline of 258 civilian calls per activation in the two days following an officer-injury-protest, suggests that a rebound in disorder offenses in the later periods may be due to a rise in underlying disorder crimes, as civilians report less and marginal offenders adjust

Table 9—Event Study Coefficients (14-Day Window, 2-Day Bins) — Full Sample

	Log Officer	Log Civilian	Log Violent	Log Disorder
Days -13	-0.0311	-0.0236	-0.0941	-0.0166
	(0.0901)	(0.0303)	(0.110)	(0.0448)
Days -12/-11	-0.0813	0.0177	0.0416	0.00783
	(0.107)	(0.0190)	(0.0688)	(0.0415)
Days -10/-9	0.0321	0.0282	0.110	0.0788*
	(0.0976)	(0.0263)	(0.102)	(0.0424)
Days -8/-7	0.0718	0.0222	-0.0995*	0.0359
	(0.0879)	(0.0264)	(0.0551)	(0.0513)
Days -6/-5	0.0372	0.0186	0.00856	0.0455
	(0.0996)	(0.0335)	(0.0581)	(0.0536)
Days -4/-3	-0.106	-0.000433	0.0224	0.0556
	(0.108)	(0.0244)	(0.0712)	(0.0423)
Days 0/1	-0.420***	-0.0923*	0.0913	-0.158***
	(0.157)	(0.0477)	(0.0764)	(0.0582)
Days 2/3	-0.204	-0.135***	0.119	-0.108*
	(0.163)	(0.0455)	(0.0740)	(0.0649)
Days 4/5	-0.214	-0.167***	0.109	-0.151*
	(0.189)	(0.0520)	(0.0993)	(0.0852)
Days 6/7	-0.308	-0.188***	0.0311	-0.151
	(0.189)	(0.0582)	(0.0816)	(0.0931)
Days 8/9	-0.245	-0.160***	0.0168	-0.145
	(0.172)	(0.0566)	(0.0700)	(0.0946)
Days 10/11	-0.135	-0.118**	0.0411	-0.0470
	(0.174)	(0.0544)	(0.0859)	(0.0990)
Days 12+	-0.248**	-0.0717***	-0.00238	-0.0206
	(0.125)	(0.0275)	(0.0407)	(0.0380)

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Interaction-weighted estimates Sun and Abraham (2021) from a 14-day event window. Adjacent pairs of event-time days are pooled into 2-day bins; the reference period is days -2 to -1. Post-period bins are reported above; pre-period bins omitted for space. City and date fixed effects absorbed. Controls as in Table 7. Standard errors clustered at the city level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10—Event Study Coefficients (14-Day Window, 2-Day Bins) — ShotSpotter Subsample

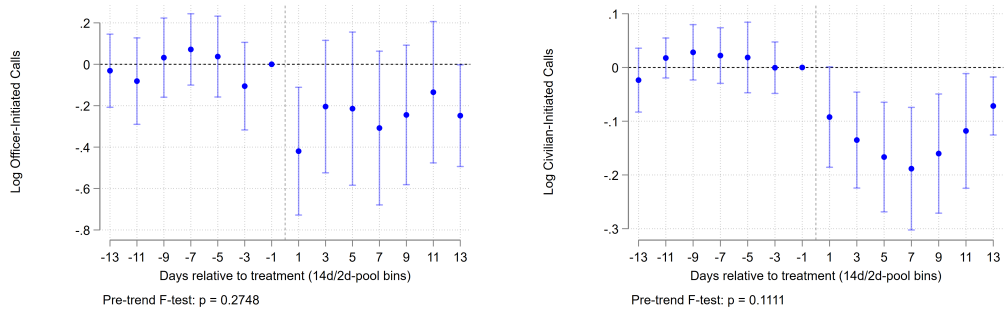
	Log Officer	Log Civilian	Log Violent	Log Disorder	Officer/Shot	Civilian/Shot
Days -13	-0.0185	-0.0115	-0.261	-0.0955	17.04	341.4
	(0.166)	(0.0444)	(0.170)	(0.0631)	(39.68)	(268.2)
Days -12/-11	-0.136	0.0166	-0.00599	-0.0458	2.510	187.6
	(0.164)	(0.0278)	(0.179)	(0.0547)	(12.81)	(127.0)
Days -10/-9	-0.120	-0.0244	-0.0805	0.000876	15.64	118.7
	(0.127)	(0.0251)	(0.135)	(0.0520)	(27.80)	(92.20)
Days -8/-7	-0.00773	-0.0366**	-0.107	-0.0853	83.86**	123.7
	(0.150)	(0.0184)	(0.108)	(0.0693)	(42.65)	(88.29)
Days -6/-5	-0.0937	-0.0249	-0.0418	-0.176***	-21.94	-33.38
	(0.224)	(0.0285)	(0.0359)	(0.0230)	(29.33)	(65.97)
Days -4/-3	-0.0516	-0.0622*	-0.103	-0.0640	-19.37	-120.2
	(0.225)	(0.0352)	(0.0708)	(0.0669)	(32.51)	(142.5)
Days 0/1	-0.341	-0.0631	0.131	-0.242***	-41.36	-257.8***
	(0.437)	(0.0728)	(0.0943)	(0.0565)	(55.00)	(79.01)
Days 2/3	0.0820	-0.111*	0.175***	-0.113**	-52.71	-155.5
	(0.447)	(0.0569)	(0.0612)	(0.0543)	(68.62)	(139.0)
Days 4/5	0.172	-0.156***	0.112	-0.0850	-36.32	28.84
	(0.435)	(0.0455)	(0.0988)	(0.0563)	(63.99)	(128.5)
Days 6/7	-0.0534	-0.173***	0.112	-0.199***	-62.71	-183.0**
	(0.366)	(0.0420)	(0.0842)	(0.0769)	(50.07)	(92.06)
Days 8/9	0.271	-0.146***	0.286*	-0.240**	-56.31	-256.5***
	(0.418)	(0.0489)	(0.165)	(0.107)	(63.41)	(51.77)
Days 10/11	0.376	-0.141*	-0.0184	-0.0222	-27.77	47.22
	(0.399)	(0.0738)	(0.167)	(0.0894)	(62.35)	(114.6)
Days 12+	0.248	-0.0921**	-0.0495	0.0278	-68.49***	-206.5***
	(0.265)	(0.0377)	(0.0868)	(0.0627)	(21.93)	(22.65)

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Same as Table 9, restricted to the 9-city ShotSpotter subsample.

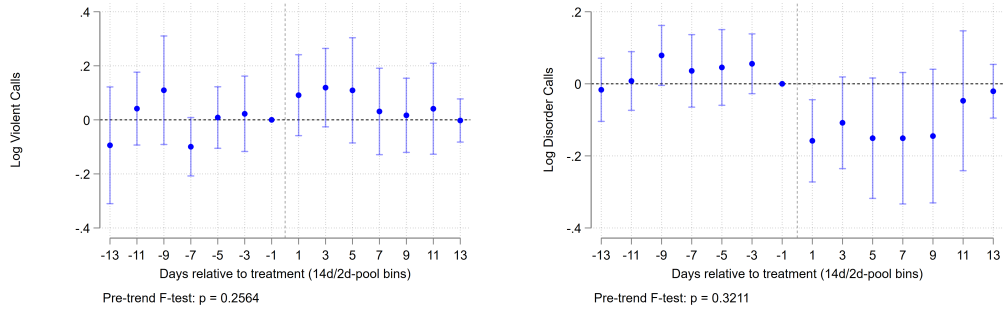
Figure 7. Dynamic Treatment Effects — Officer- and Civilian-Initiated Calls, Full Sample (14d/2d)



(a) Officer-Initiated Calls (b) Civilian-Initiated Calls

Note: Sun-Abraham (2021) interaction-weighted event study estimates. Window: 14 days, 2-day bins. The reference period is days −2 to −1, omitted because 2-day pooling requires the two to be dropped together. Bars are 95% confidence intervals clustered by city. Covariates include weekly COVID-19 cases, daily temperature, precipitation, and Google mobility indices. Treated cities (17) are those where a documented BLM protest with officer injuries occurs. Control (23) cities are never-treated. Manchester, NH has no CAD data after November 19, 2020 and so leaves the comparison group for later calendar dates; this affects only the open-ended end-cap coefficient.

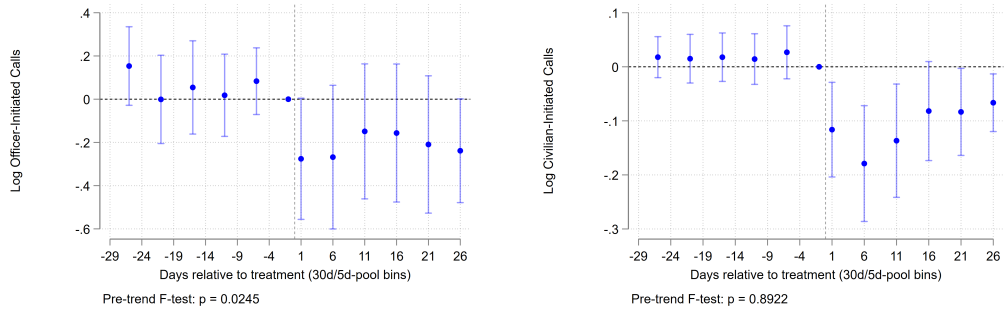
Figure 8. Dynamic Treatment Effects — Violent and Disorder Calls, Full Sample (14d/2d)



(a) Violent Calls (b) Disorder Calls

Note: See notes to Figure 7.

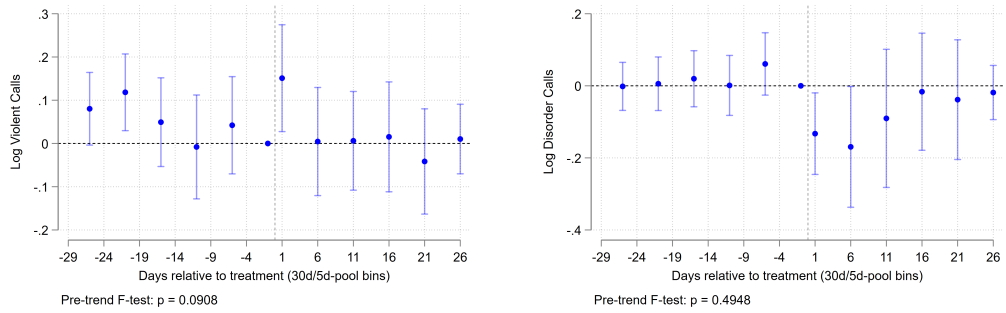
FIGURE 9. DYNAMIC TREATMENT EFFECTS — OFFICER- AND CIVILIAN-INITIATED CALLS, FULL SAMPLE (30D/5D)



(a) Officer-Initiated Calls (b) Civilian-Initiated Calls

Note: See notes to Figure 7. Window extended to 30 days with 5-day bins.

FIGURE 10. DYNAMIC TREATMENT EFFECTS — VIOLENT AND DISORDER CALLS, FULL SAMPLE (30D/5D)



(a) Violent Calls (b) Disorder Calls

Note: See notes to Figure 9.

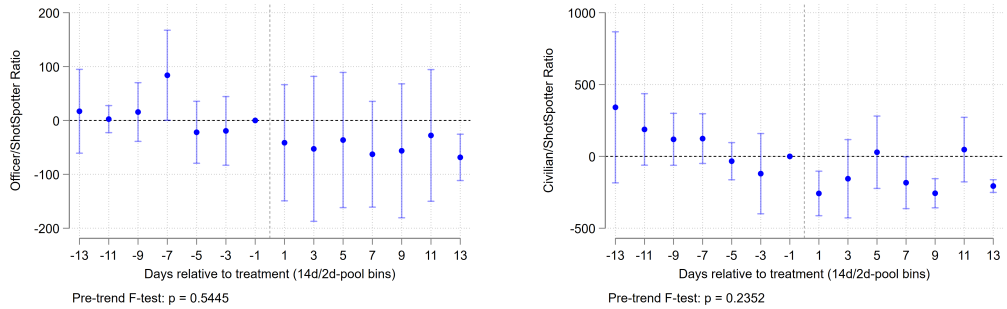
to a reduced probability of being caught. Violent calls rise immediately after a protest, but not significantly so, and remain stable throughout the sample.

The main outcome of interest, officer-initiated calls, falls sharply following treatment (roughly 42% in the two days following a protest) and point estimates remain persistently negative throughout the 14-day post-treatment window. In the post-14-day window, officer-initiated calls remain significantly lower than their pre-treatment level, suggesting that depolicing occurs post-treatment.

C. Conclusion and Policy Implications

This paper provides evidence that exposure to protest-related outrage, identified by officer injury at post-Floyd BLM protests, causes a significant and durable reduction in both proactive policing and civilian engagement with police. Officer-initiated call rates fall and remain lower throughout the post-treatment win-

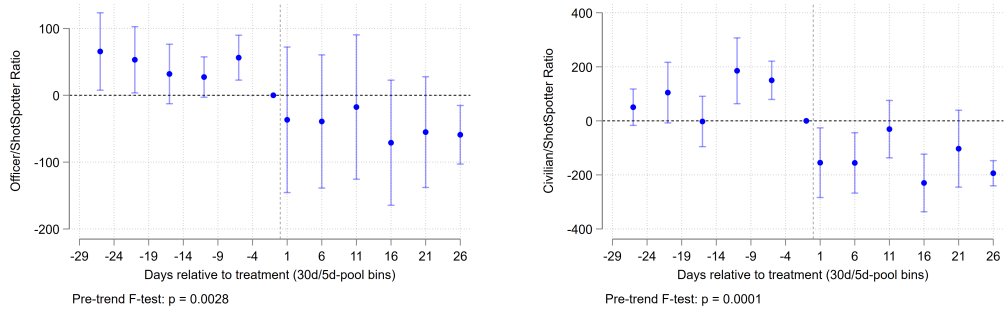
FIGURE 11. DYNAMIC TREATMENT EFFECTS — CALL-TO-SHOT RATIOS, SHOTSPOTTER SUBSAMPLE (14D/2D)



(a) Officer/Shot Ratio (b) Civilian/Shot Ratio

Note: Sun–Abraham (2021) interaction-weighted event study estimates. Window: 14 days, 2-day bins. The reference period is days −2 to −1, omitted because 2-day pooling requires the two to be dropped together. Bars are 95% confidence intervals clustered by city. Covariates include weekly COVID-19 cases, daily temperature, precipitation, and Google mobility indices. Restricted to the 9 cities with usable ShotSpotter coverage; Fort Lauderdale is excluded because its activations begin only in October, and Menlo Park because it records activations on just six days in 2020. Treated cities (4) are those where a documented BLM protest with officer injuries occurs. Control (5) cities are never-treated. Outcomes are call-to-shot ratios, dividing officer- or civilian-initiated calls by ShotSpotter activations, each incremented by 1 to avoid division by zero.

FIGURE 12. DYNAMIC TREATMENT EFFECTS — CALL-TO-SHOT RATIOS, SHOTSPOTTER SUBSAMPLE (30D/5D)



(a) Officer/Shot Ratio (b) Civilian/Shot Ratio

Note: See notes to Figure 11. Window extended to 30 days with 5-day bins.

dow, while civilian-initiated calls decline in a pattern consistent with reduced discretionary reporting rather than a fall in underlying crime. The ShotSpotter subsample results strengthen this interpretation: a decline in civilian calls per gunshot activation suggests that marginal reporters in treated cities avoid contacting police following outrage, independent of any shift in gun crime. A decline in officer-initiated calls per activation strengthens the depolicing evidence.

These findings contribute to the literature on depolicing and the police-public relationship in two distinct ways. First, they confirm that depolicing occurs in the context of the largest instance of public backlash against law enforcement in U.S. history. Second, the simultaneous decline in civilian reporting and officer-initiated calls underscores the difficulty of interpreting reported crime changes following episodes of public outrage: both channels move in directions that suppress reported crime, potentially masking underlying shifts in criminal activity.

The policy implications are clear. Deterioration of the police-public relationship following high-profile protests reduces the capacity of law enforcement to prevent crime through both supply and demand channels. Officers reduce discretionary engagement, and civilians reduce discretionary reporting. Departments seeking to mitigate depolicing effects following outrage episodes may benefit from proactive communication strategies that maintain officer morale and community trust simultaneously, rather than treating depolicing and non-reporting as independent problems.

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